Highlights

Bilevel Optimization for Supply Chain Network Design under Carbon Taxation

- Proposes a bilevel model integrating government carbon tax and supply chain design decisions.
- Balances environmental impact (leader) and cost minimisation (follower) in a closed-loop network.
- Develops a deterministic bounding procedure with convergence guarantee for bilevel MIP.
- Reproducible artifacts validate reported DBP solutions and saved follower decisions.